BANK ANALYTICS PROJECT

COMPARATIVE ANALYSIS USING

Excel

Power BI

Tableau

SQL

Presented by: Suhas P S

INTRODUCTION

- ❖ The banking dataset used in this project represents key operational aspects of a financial institution, including **customer information, loan details, branch data, product types, and repayment records**. This dataset provides a structured view of how banking services are delivered to customers and how financial performance can be measured at different levels such as client, branch, and product.

- ❖ The purpose of analyzing this banking dataset is to gain insights into **loan disbursement patterns, repayment behavior, and overall banking performance**. By examining relationships between clients, loans, branches, and products, the dataset enables a detailed understanding of how different banking units contribute to business outcomes.

PROJECT OBJECTIVE

- To gain a comprehensive understanding of **bank transaction patterns** across time and branches.
- To measure the bank’s **financial performance** using well-defined KPIs.
- To identify **transaction irregularities** and potential risk indicators.
- To evaluate **customer engagement** through transaction frequency analysis.
- To enable **branch-level performance comparison** for operational insights.
- To present insights through **clear, interactive dashboards** that assist management in decision-making.

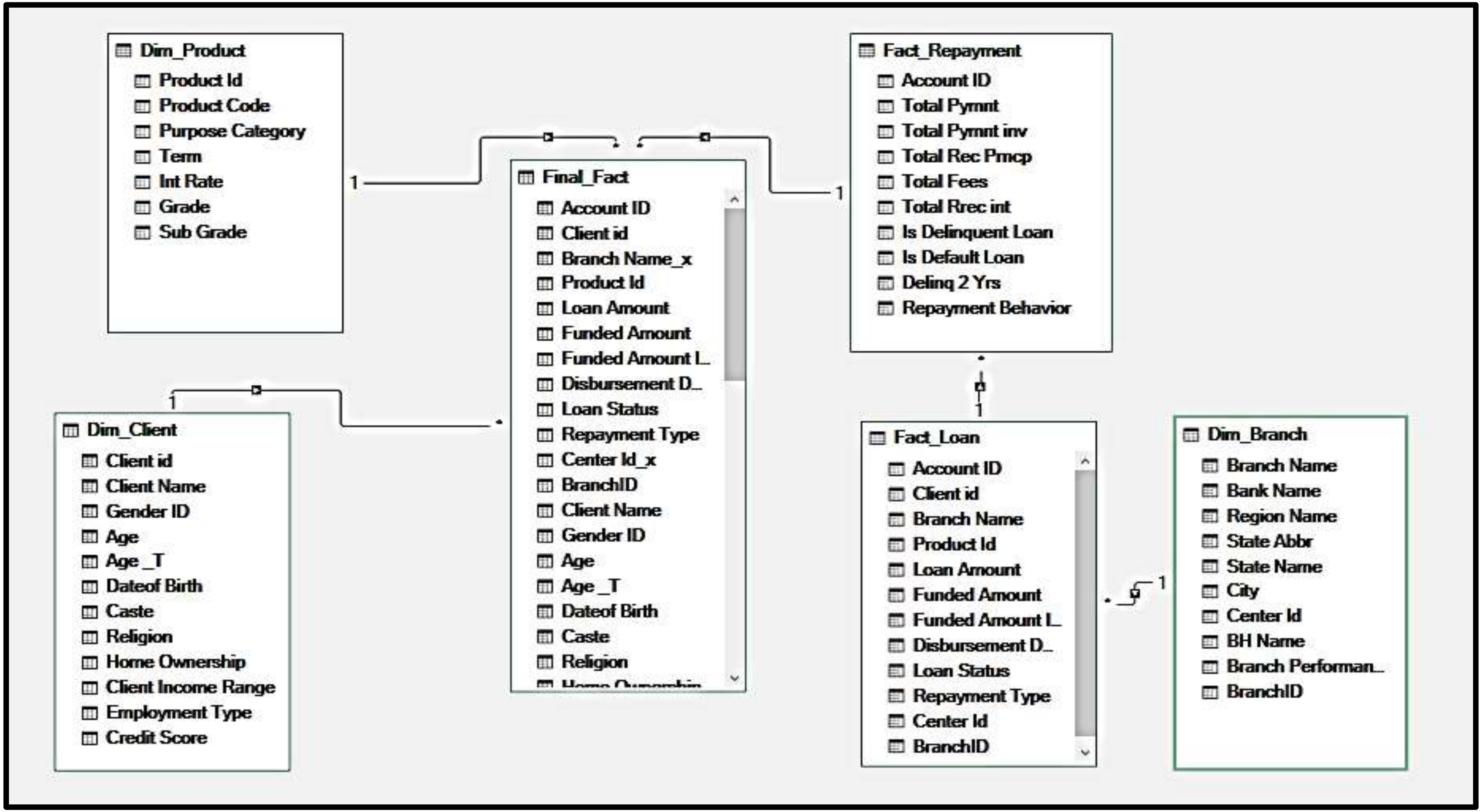
WHY THIS PROJECT

- The banking and financial services industry plays a crucial role in the economy and generates large volumes of structured data related to customers, loans, branches, and financial transactions. Analyzing this data effectively is essential for improving operational efficiency, managing credit risk, and supporting strategic decision-making. This motivated us to select a **Banking Data Analytics project** as it represents a real-world, data-intensive business scenario.

- This project was selected to gain practical exposure to **loan performance analysis, repayment behavior, and risk identification**, which are key challenges faced by banks. The dataset provides a comprehensive view of banking operations, making it ideal for understanding how different factors such as branch, product, and customer behavior influence overall performance.

- Overall, this project was selected because it provides a **practical, business-oriented learning experience** that closely reflects real banking analytics use cases and prepares us for real-world data analyst roles.

DATA MODELLING FRAMEWORK

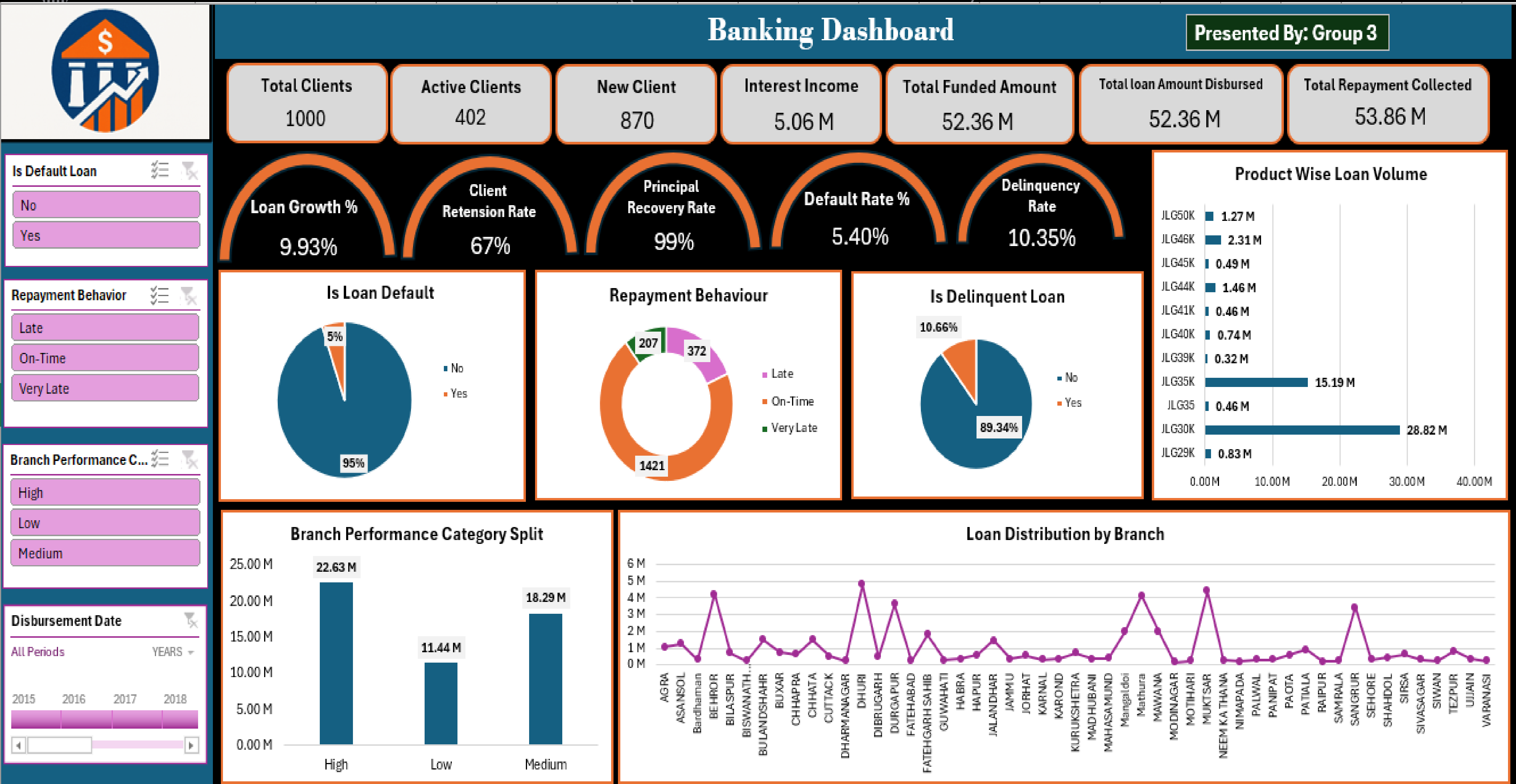


KPI’S

Total Clients: 1000	Represents the total number of customers currently associated with the bank. This indicates the overall customer base size.
Client Retention Rate: 67%	Measures the percentage of customers who continue their relationship with the bank. A 67% retention rate suggests a moderately strong customer loyalty level.
Total Loan Amount Disbursed: 52M	Represents the total value of loans issued by the bank, reflecting lending activity and business expansion.
Loan Growth % : 16.18%	Reflects the growth in loan disbursement compared to the previous period, indicating positive business momentum.
Principal Recovery Rate: 99.05%	Measures the percentage of principal amount successfully recovered. A very high value indicates excellent loan recovery efficiency.

Default Rate: 5%	**Shows the percentage of loans that have defaulted. A relatively low value suggests effective credit risk management.**
Delinquency Rate: 10.35%	Measures the proportion of loans with delayed payments. This helps identify potential future default risks.
Performance Category Split: • **Low Risk: 43%** • **Medium Risk: 35%** • **High Risk: 22%**	This distribution highlights that most loans fall under low to medium risk, indicating a balanced and controlled loan portfolio.
Year-Over-Year Growth %	• Shows fluctuations in growth across years • Strong positive growth in recent years (up to **16.18%**) indicates recovery and expansion after a decline period

Excel Dashboard

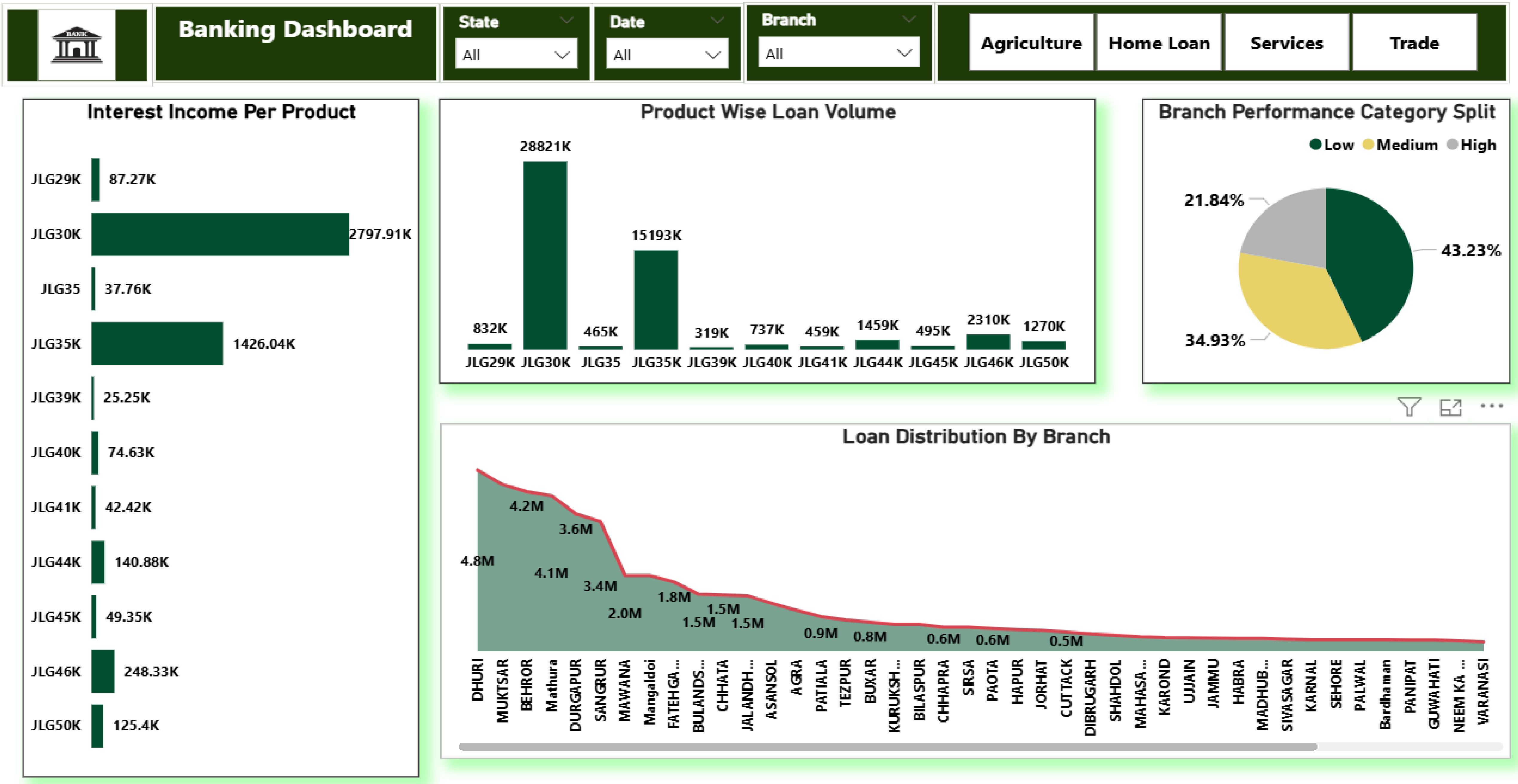


Power Bi Dashboard - 1

Banking KPI

1K	Total Clients
402	Active Clients
34	New Clients
67.24	Client Retention Rate
52M	Total Loan Amount Disbursed
52M	Total Funded Amount
26.18K	Average Loan Size
54M	Total Repayments Collected
99.05%	Principal Recovery Rate
5.00%	Default rate
10.35%	Delinquency Rate
71.05%	On-Time Repayment Behaviour%
5.06M	Interest Income

Power Bi Dashboard - 2



Power Bi Dashboard - 3

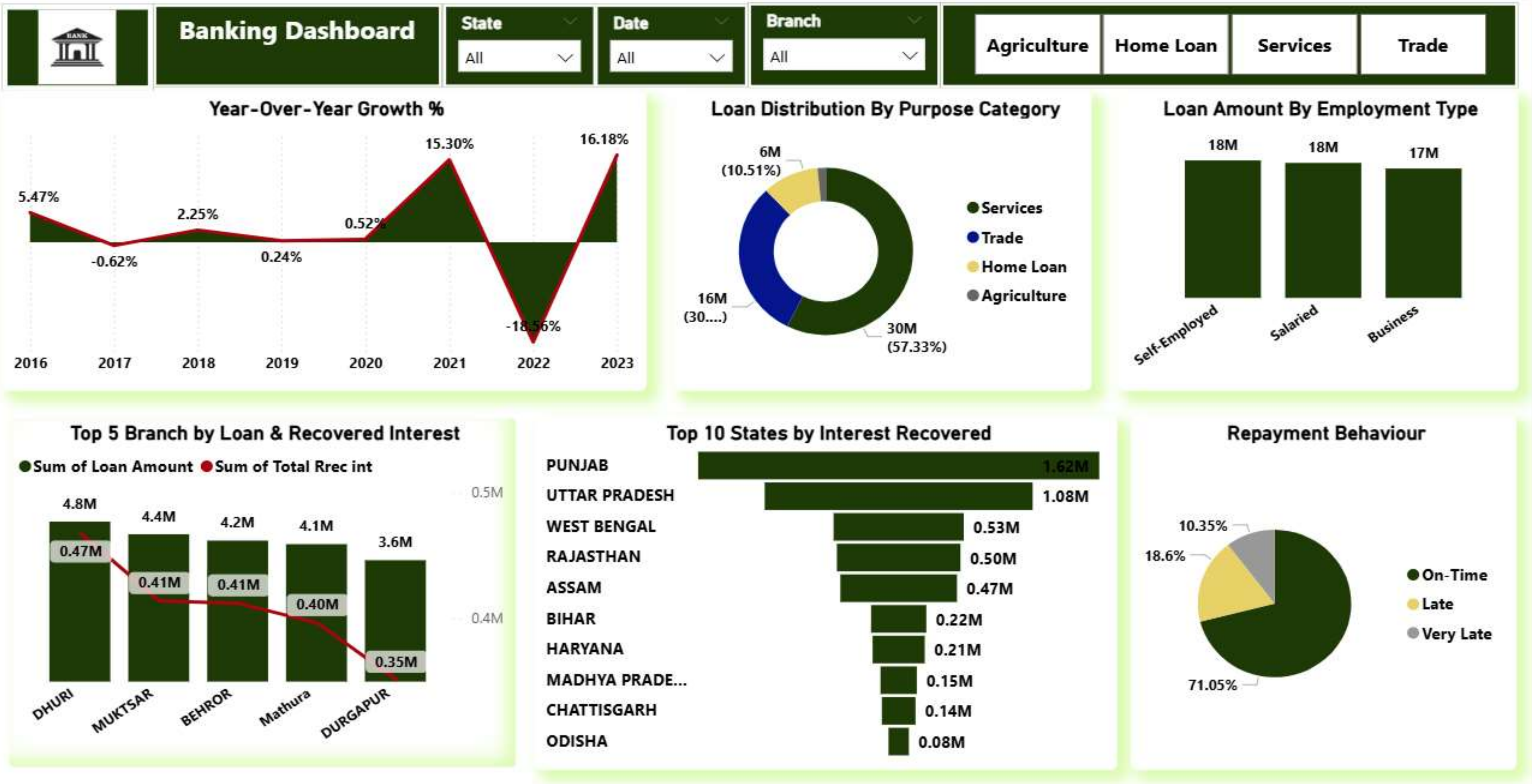
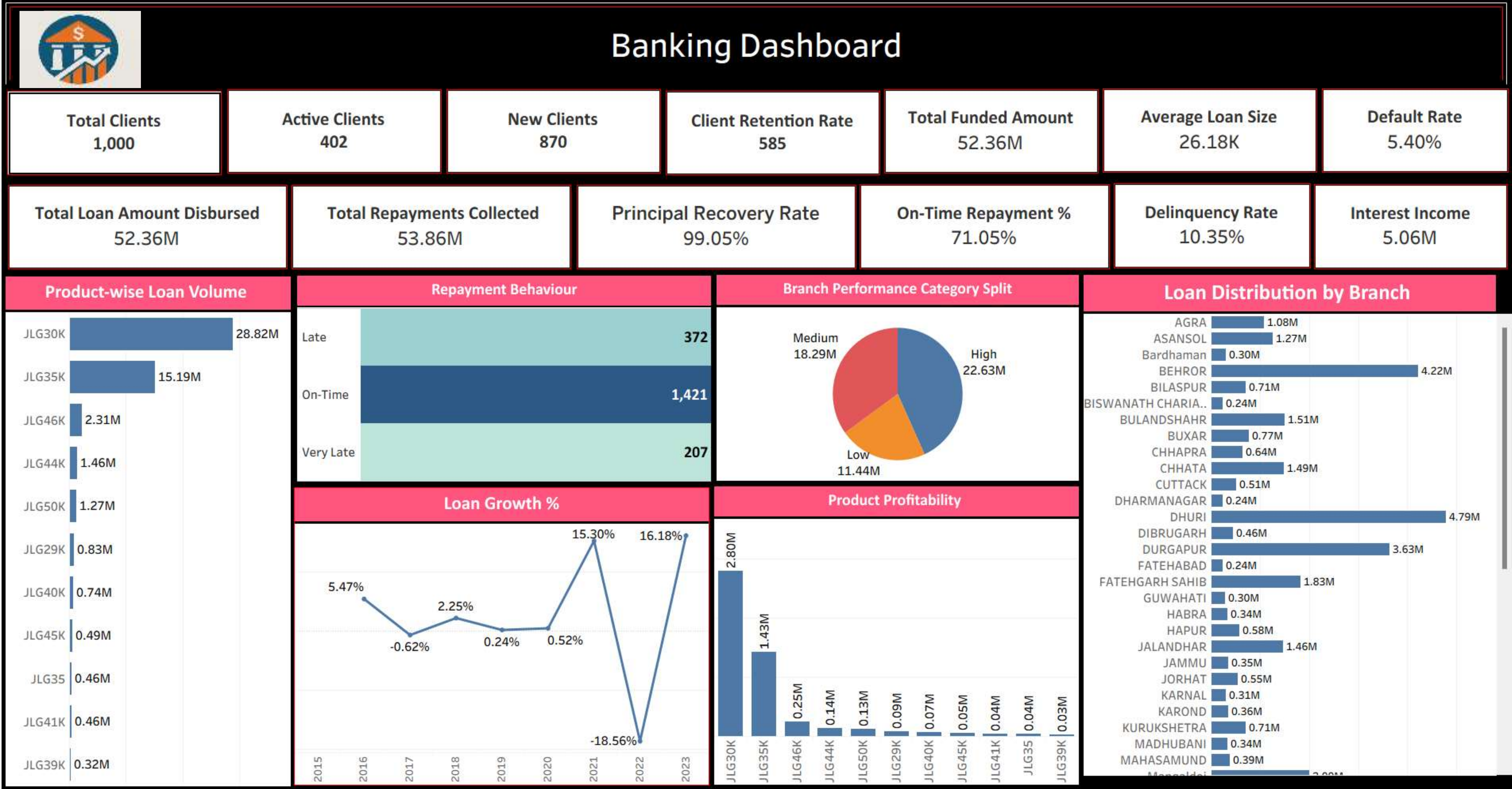


Tableau Dashboard



SQL QUERIES

```sql
Create Database Bank_Analytics;
Use Bank_Analytics;
Alter table final_fact
Modify column `Disbursement Date` date;
Show tables;
-- 1. TOTAL CLIENTS
select count(`client id`) from dim_client as Total_Clients;

-- 1. ACTIVE CLIENTS
select count(`Loan Status`) from  final_fact where `Loan Status` = "Active";

-- 2. NEW CLIENTS
select count(`Client Id`) from final_fact
where Year(`Disbursement Date`) not in ("2015","2016","2017","2018","2019","2020","2021","2022");

```

SQL QUERIES

```
-- 3. Client Retention Rate
 select ROUND(
        100 * SUM(transaction_count > 1) / COUNT(*),
        2
    ) AS Client_Retention_Rate
FROM (
    SELECT
        `Client id`,
        COUNT(*) AS transaction_count
    FROM final_fact
    GROUP BY `Client id`
) t;


-- 4.   Total Loan Amount Disbursed
select sum(`Loan Amount`) from fact_loan;

-- 5. Total Funded Amount
select sum(`funded amount`) from fact_loan;
```

SQL QUERIES

```
-- 6. Average Loan Size
select avg(`Loan Amount`) from fact_loan;

-- 7. Loan Growth %
With Yearly_Sales as(
select Year(`Disbursement Date`) as Year, sum(`Loan Amount`) as Amount from final_fact
group by Year(`Disbursement Date`))
select Year, Amount, Lag(Amount) over (order by year) as Previous_Year,
((Amount - Lag(Amount) over (order by year) )/Lag(Amount) over (order by year))*100 as Growth_Percentage
 from Yearly_Sales order by Year desc ;

-- 8. Total Repayments Collected
Select sum(`total pymnt`) from final_fact;

-- 9. Principal Recovery Rate
select (sum(`Total Rec Prncp`)/sum(`Loan Amount`)* 100.00 )from final_fact;
```

SQL QUERIES

```sql
-- 10. Interest Income
select sum(`Total Rrec int`) from fact_repayment;

-- 11. Default Rate
Select (sum(Case When `Is Default Loan` ="Y" Then 1 Else 0 END) *100.00 / count(*)) as default_range_Percentage
from final_fact;

-- 12. Delinqquncy Rate
Select(sum(Case When `Is Delinquent Loan`="Y" Then 1 Else 0 END)*100.00/count(*))
 as DelinquencyRate from final_fact;

-- 13. On time Repayment %
Select (sum(Case When `Repayment Behavior`="On-Time" then 1 Else 0 End) /count(*))*100.00
as On_Time_Repayment from final_fact;

-- 14. Loan Distribution By Branch
select `Branch Name_x`,sum(`Loan Amount`) from final_fact group by `Branch Name_x` order by sum(`Loan Amount`) desc;
```

```sql
-- 16. Product wise Loan Volume
SELECT `Product Id`, sum(`Loan Amount`)  as Amount from final_fact group by `Product Id`;

-- 17. Interest Income Per Product
Select `Product Id`, sum(`Total Rrec int`) as Interest_Recovered
from final_fact group by `Product Id`
Order by sum(`Total Rrec int`) desc;
```

PERFORMANCE ANALYSIS

- ❖ The banking performance analysis evaluates customer behavior, loan portfolio strength, repayment efficiency, and risk indicators using dashboard-based insights.

- ❖ The bank serves **1,000 clients**, but only **402 are active**, indicating low customer engagement. With **242 new clients** and a 67% retention rate reminding that most business comes from existing customers rather than new active participation.

- ❖ The **total loan amount disbursed is ₹52.36M** with an **average loan size of ₹26.18K**, showing a focus on small-ticket lending. Loan growth stands at **16.18%**, but year-over-year trends reveal volatility, including a sharp decline in 2022, highlighting sensitivity to market conditions.

- ❖ Repayment performance is strong, with **₹53.86M collected** and a **99% principal recovery rate**. **71.05% of repayments are on time**, ensuring healthy cash flow. However, the **10.35% delinquency rate** signals early repayment stress, even though the **default rate remains low at around 5%**.

- ❖ Branch and product analysis shows uneven performance, with a few branches and products (JLG20K and JLG25K) contributing the majority of loan volume and interest income. This

TREND & TIME BASED ANALYSIS

- The trend and time-based analysis highlights how the bank's performance changes across years, months, and repayment cycles.

- The **Year-Over-Year (YoY) growth trend** shows moderate growth in most years, with steady increases up to 2021. However, there is a **sharp decline in 2022**, indicating the impact of external or operational challenges, followed by a **strong recovery in 2023 (16.18%)**, reflecting improved lending activity and business stabilization.

- Monthly transaction trends indicate relatively **consistent transaction volumes** throughout the year, suggesting stable customer activity. Any minor fluctuations point toward seasonal effects rather than structural issues.

- From a repayment perspective, **on-time repayment (71.05%)** remains stable over time, supporting healthy cash flows. However, the presence of a **10.35% delinquency rate** suggests that repayment behavior requires continuous monitoring, especially during economic stress periods.

KEY INSIGHTS

Client Base Overview

The bank manages **1,000 total clients**, out of which only **402 are active.** This indicates that nearly **60% of customers are inactive**, highlighting a major opportunity for customer reactivation.

Credit Risk Overview

The **default rate is 5.40%**, meaning **95% of loans are non-default**. This shows strong credit assessment and lending discipline across the portfolio.

Delinquency Warning Signal

The **delinquency rate is 10.35%**, which is almost double the default rate. This suggests early repayment stress that could turn into future defaults if not addressed.

Repayment Behaviour Analysis

Out of total repayments, **1,421 are on-time**, while **372 are late** and **207 are very late**. The dominance of on-time payments reflects healthy borrower repayment behavior.

Recovery Strength

The **principal recovery rate is an impressive 99%**, showing excellent loan recovery efficiency. This minimizes losses and strengthens the bank's cash flow position.

Disbursement vs Collection

Total loan amount disbursed is **₹52.36M**, while total repayment collected is **₹53.86M**. Collections exceeding disbursements indicate strong recovery from previous loans.

Branch Performance Distribution

High-performing branches contribute **₹22.63M**, compared to **₹18.29M (Medium)** and **₹11.44M (Low)**.
Performance gaps suggest uneven operational efficiency across branches.

Product Loan Concentration

Two products dominate loan volume:

- **JLG30K – ₹28.82M**
- **JLG35K – ₹15.19M**

This shows high dependence on a few loan products, increasing concentration risk.

STRATEGIC RECOMMENDATIONS

Shift Focus from Loan Volume to Customer Acquisition

With only 402 active clients out of 1,000, growth is clearly not acquisition-led. The bank should prioritize onboarding new active borrowers, not just increasing loan sizes for existing customers.

Introduce Early-Warning Systems for Delinquency

A **10.35% delinquency rate**, nearly double the **5.4% default rate**, signals early repayment stress. Automated alerts for missed EMIs can prevent delinquency from converting into defaults.

Stabilize Year-on-Year Loan Growth

Loan growth shows **high volatility**, including a **−18.56% drop in 2022**. The bank should adopt conservative, steady lending targets to avoid boom-and-bust cycles.

Reduce Product Concentration Risk

Over **80% of loan volume** is concentrated in two products (**JLG30K & JLG35K**). Expanding mid-tier loan products will reduce dependency risk and improve portfolio resilience.

Reactivate Inactive Customers

Nearly **60% of customers are inactive**, representing untapped potential. Targeted re-engagement campaigns can generate low-cost growth compared to acquiring new customers.

Strengthen Delinquent Loan Management

Since delinquent loans exceed defaulted loans, focused recovery teams should address **late and very-late payers** before accounts deteriorate further.

Replicate High-Performing Branch Practices

High-performance branches generate **₹22.63M**, double that of low performers. Best practices from top branches should be standardized across weaker branches.

Build Growth Metrics Beyond Averages

The **9.93% loan growth KPI** hides year-wise instability. Management should track **median growth** and **risk-adjusted growth metrics** instead of simple averages.

CONCLUSION

The analysis shows that the bank has **strong financial stability**, supported by high repayment discipline, a **99% recovery rate**, and controlled default levels. However, a large inactive customer base, higher delinquency compared to defaults, and concentration in a few products and branches highlight key areas for improvement. Focusing on **customer activation, early risk monitoring, and diversification** will help the bank achieve more consistent and sustainable growth.

THANK YOU